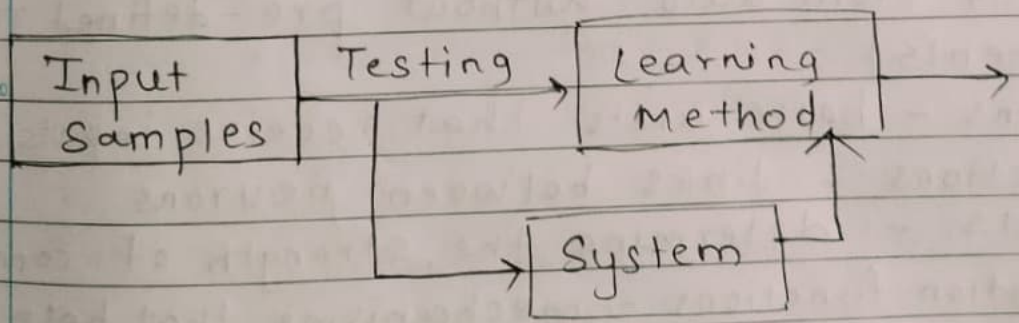
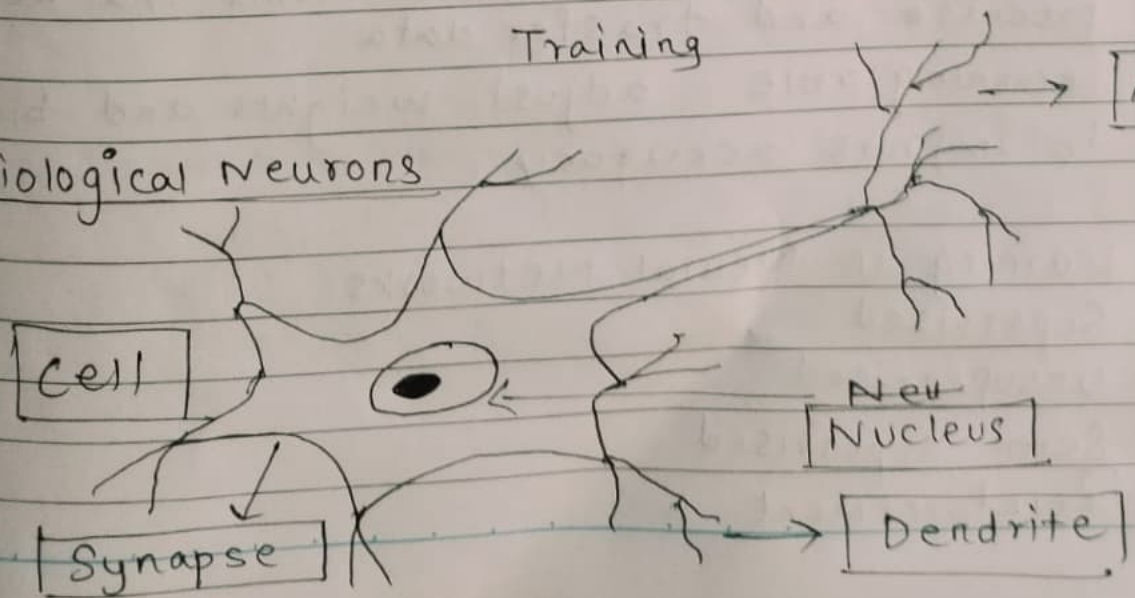
Mod 1

	Machine Learning	Deep Learning
1.	superset of DL	subset of ML
2.	uses structured data	uses neural networks
3.	evolution of AI	evolution of ML
4.	thousands of data points	millions of data points
5.	Performed using CPU	Performed using GPU
6.	More human intervention	Less human intervention
7.	Feature engineering	Not required

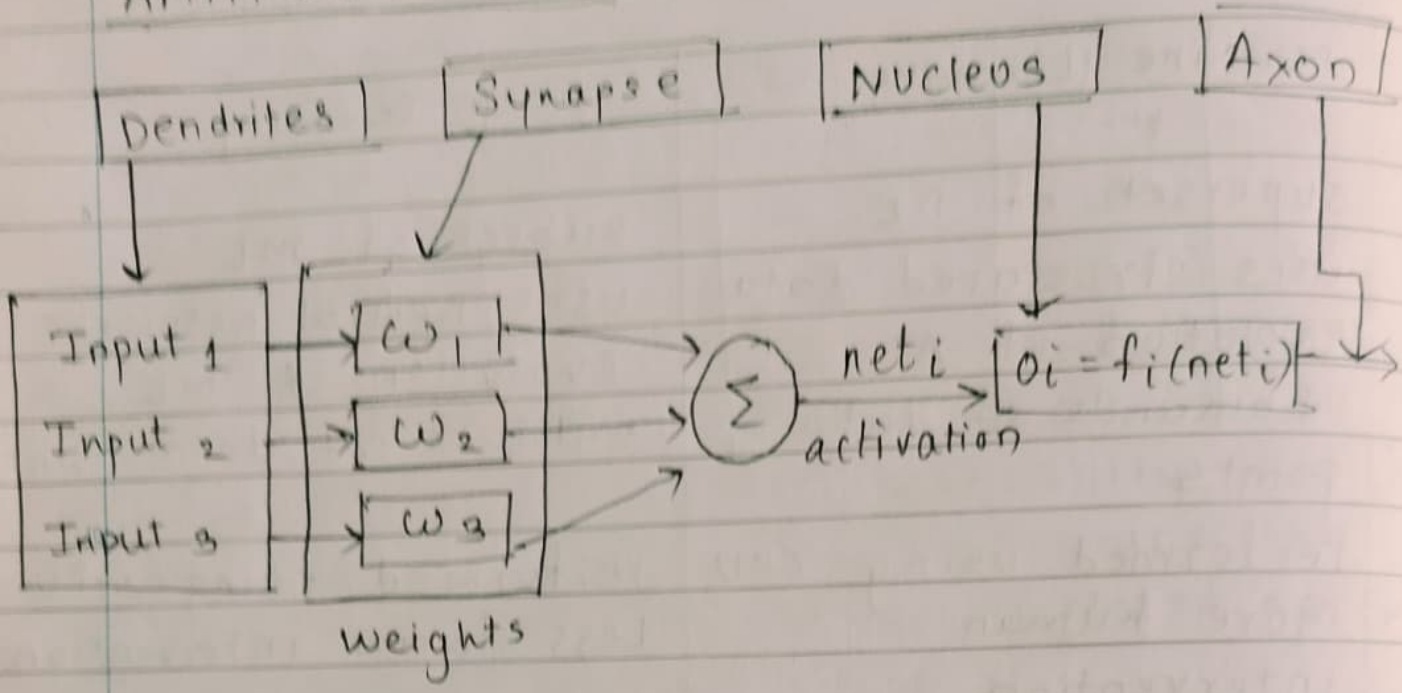
Deep learning uses neural networks with multiple layers to analyze complex patterns and relationships in data.



Biological Neurons



Artificial Neurons



Neural Networks

Neural networks are models that mimic the complex functions of human brain. They are capable of learning and identifying patterns directly from data without pre-defined rules.

Components

1. Neurons - basic units that receive inputs.
2. Connections - links between neurons
3. Weights - determine the strength of connection
4. Activation functions - mechanisms that help process and trasfer data
5. Learning rule - adjust weights and biases to improve accuracy.

Learning in Neural Networks

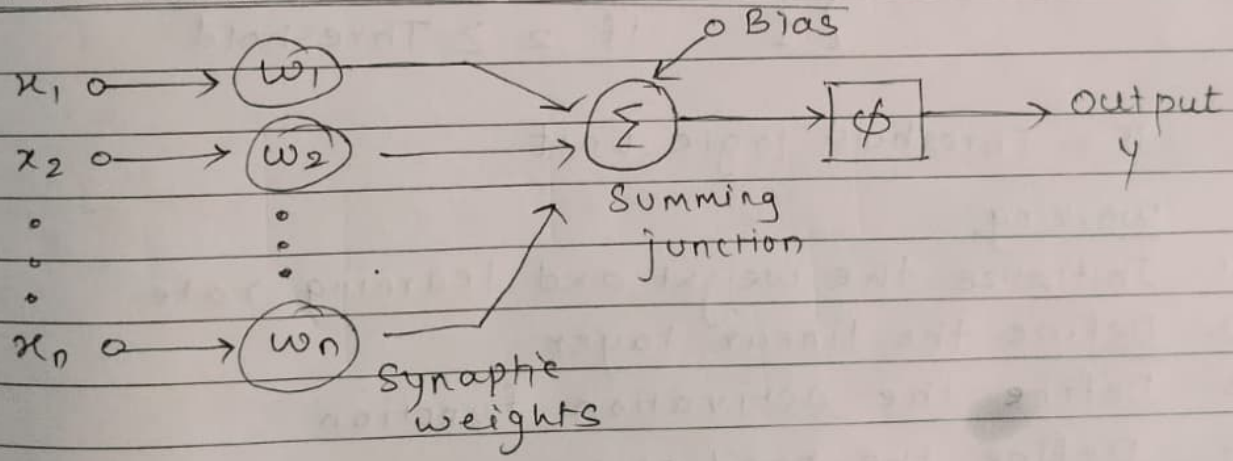
1) Supervised
2) Unsupervised
3) Semi-supervised
4) Reinforcement

1) Supervised - training data includes desired outputs
2) Unsupervised - training data does not include desired outputs
3) Semi-supervised - training data includes a few desired outputs
4) Reinforcement - rewards from a sequence of actions

- **Factors affecting performance.**

1) Type of training
2) Type of feedback
3) Learning algorithms

- **McCulloch Pitt's Neuron Model**



$x \cdot w = x_1 w_1 + x_2 w_2 + \dots + x_n w_n$

If $x \cdot w \geq \phi$ (threshold value)

output = 1

else

output = 0

let ϕ be 2

x_1	x_2	$x_1 + x_2$	out
1	1	2	1
1	0	1	0
0	1	1	0
0	0	0	0

- Perceptron.

simplest neural network architecture
Perceptron performs binary classification that maps input features to an output decision, usually into binary values 0 or 1.

1) Single-layer perceptron → limited to learning linearly seperable patterns.

2) Multi-layer perceptron → possess enhanced processing capabilities, adept at handling more complex patterns

$$z = w_1x_1 + w_2x_2 + \dots\dots\dots + w_nx_n = X^T W$$

$$h(z) = \begin{cases} 0 & \text{if } z < \text{Threshold} \\ 1 & \text{if } z \geq \text{Threshold} \end{cases}$$

T = Threshold logic unit

working

1. Initialize the weight and learning rate
2. Define the linear layer.
3. Define the activation function
4. Define the prediction
5. Define the loss function
6. Define the training
7. Fit model.

Date :

• X-OR Problem

It is a simple logic gate problem that can be solved by neural networks.

Truth Table for XOR

Input A	Input B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

The problem cannot be solved by a perceptron as the data is not linearly seperable..

Solution

1) Input to hidden layer transformation.

$h_1 = \sigma(w_{11} A + w_{12} B + b_1)$

$h_2 = \sigma(w_{21} A + w_{22} B + b_2)$

where $\sigma(x) = \frac{1}{1 + e^{-(x)}}$ (activation function)

2) Hidden to output

$output = \sigma(w_{31} h_1 + w_{32} h_2 + b_3)$

3) Learning weights and biases.

